Automated Scanning & Dosing Subsystem

MAGTRACK magnetic wall-climbing pipeline-inspection robot · Capstone Design Project II, Higher Colleges of Technology
Designed by Elham Mohammed Kedir (sole designer) · Rev 2, 11 October 2026 · SolidWorks

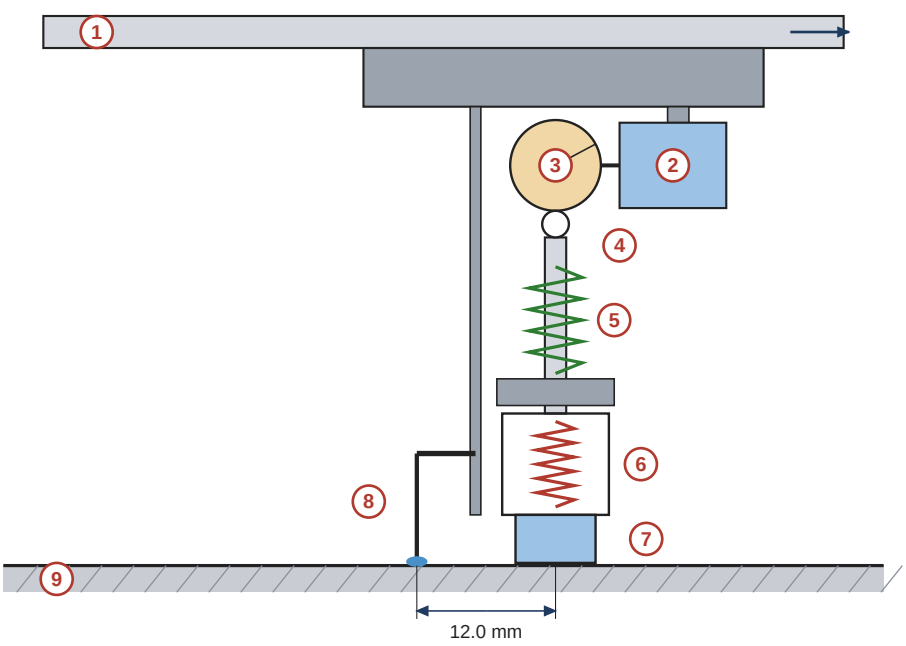
1. Problem

The first design dragged a spring-loaded ultrasonic probe over corroded steel while couplant pumped continuously. Three problems followed:

- **Probe wear:** dragging abrades the transducer face and degrades the echo.
- **Couplant loss:** continuous pumping runs off vertical walls and empties the 250 mL tank early.
- **Poor scan control:** with an 8.0 mm probe the helix pitch must stay under 7.2 mm. On a DN1000 pipe that means holding the heading to about 0.13°, and a 0.5° drift gives roughly 28 mm of pitch error.

2. Concept

Replace continuous drag with **indexed, step-and-shoot scanning**: the crawler locks magnetically, a lead-screw stage moves the probe along the pipe axis in 6 mm steps, a cam lowers the probe at each stop, and a micro-pump places a couplant bead 12 mm ahead so it is in place when the probe arrives.



Key

1. X-axis stage (Nema 11, 200 mm)
2. MG995 servo (drives cam)
3. Single-lobe cam, 14 mm stroke
4. Roller follower rod, Ø8 mm
5. Return spring, 1.07 N/mm
6. Compliance spring, 0.54 N/mm
7. Ultrasonic probe, Ø8 mm, 5 MHz
8. Dosing needle, 12 mm ahead
9. Pipe wall

Figure 1. Side view of the stage, cam and probe assembly (schematic, not to scale).

3. Key design parameters

Item	Value	Basis
Scan grid	6.0 × 6.0 mm square, 2.0 mm overlap for an 8.0 mm probe	Pitch must be at most 0.9 × beam diameter (7.2 mm)
X-axis stage	200 mm stroke, 188 mm valid, 32 nodes per strip	First two nodes prime the needle offset
Cam	14 mm harmonic rise (90°/60°/90°/120° regions), peak pressure angle about 29°	In-line roller follower; servo reverses, so the fall flank is unused
Servo	MG995; peak demand about 0.25 N·m against 0.83 N·m rated; near zero torque in dwell	Virtual-work torque check; slow ramp through contact
Contact force	4.4 N (target 4.0–4.6 N)	Compliance spring 0.54 N/mm with 2.6 mm spacer
Return spring	1.07 N/mm, about 16 N at full throw	304 stainless wire; stress and solid-height checked
Couplant	80% glycerin, about 50 µL per node	Resists evaporation in UAE sun; pumpable
Couplant film	About 5 µm after a 0.3 s settle	Stefan squeeze-film model; echo-to-echo gauge mode removes film bias
Throughput	About 1.2–2.0 s per node; about 100 strips in 100 min	Design estimate, to be measured
Couplant supply	250 mL gives about 4,500 doses (about 140 strips)	After a 10% reserve

4. Worked calculations

Contact force. Compliance spring (304 stainless, d = 0.5 mm, D = 5.5 mm, n = 6, G = 69 GPa): $k = G \cdot d^4 / (8 \cdot D^3 \cdot n) \approx 0.54$ N/mm. With a 2.6 mm spacer and 5.5 mm over-stroke, $F = 0.54 \times (2.6 + 5.5) \approx$ **4.4 N**.

Scan tolerance. Pitch ≤ 0.9 × 8.0 mm = 7.2 mm. On DN1000 (circumference about 3192 mm) the heading angle is arctan(7.2 / 3192) ≈ 0.129°, and a 0.5° drift changes the pitch by 3192 × tan 0.5° ≈ 28 mm. This is why indexed scanning replaced a continuous helix.

5. Operating sequence and safety interlocks

Per node: **deploy** (slow through contact) → **dose** → **settle** (at least 300 ms) → **read** → **raise** to a 4 mm hop → **step** 6 mm. At the end of a strip the probe stows, the stage returns home and the crawler indexes 6 mm.

ID	Interlock
I1	The stage moves only when the CLEAR sensor reports the probe at least 3 mm off the wall (hardware gate on the stepper enable plus a firmware check)
I2	Crawler drives run only with the servo parked, CLEAR true and the stage at home, through safety relay K1 (opens on reset or power loss)
I3	If CLEAR is not confirmed within 500 ms of a retract, retry once, then fault and lock until an operator resets
I4	No pump pulse while the servo moves, and no servo move during a pump pulse
I5	The probe lowers only when the carriage is stationary

6. Verification plan

Test	What it checks	Pass criterion
1 Dose calibration	50 pulses, weighed, at 25 °C and 45 °C	Mean within ±10% of target, CV below 15%
2 Contact force and servo load	Force against servo angle; spacer selection	4.0–4.6 N nominal, 3.5–5.2 N across ±1 mm; dwell current near idle
3 Ultrasonic accuracy	Steel step wedge, 20 automated cycles per step	±0.05 mm clean, ±0.10 mm corroded
4 Coverage and cycle time	5+ strips with an ink-footprint check	Spacing at most 7.2 mm, index error within ±0.5 mm
5 Droplet retention	Bead transfer at 0/90/180/270°, 20 beads per dose	20 of 20 transferred, within ±1.5 mm, run-off at most 1 mm at 10 s
6 Spring-only return and interlocks	Servo disconnected; blocked follower; reset during a strip	Full return in every orientation; stage blocked when CLEAR is false

7. Status and open items

Design Rev 2 is complete. All six tests are planned and no results are recorded yet. Still to confirm: the transducer element diameter (sets the step size), the MG995 pulse-to-angle calibration, the cam material (nylon, PETG-CF or aluminium, because PLA is marginal at about 45 MPa contact stress), and the CLEAR sensor type and trip height.